BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MULTAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 2461 / 2462

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

BIOLOGY PAPER-I GROUP-I (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2461

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. An aphid that attacks walnut tree is being controlled biologically by:

(A) Housefly
(B) Honey bee
(C) Mosquito
(D) Wasp

Q2. Cotton is the pure form of:

(A) Cellulose
(B) Amino acid
(C) Glycogen
(D) Starch

Q3. An enzyme with its co-enzyme removed is designated as:

(A) Holoenzyme
(B) Apoenzyme
(C) Co-factor
(D) Activator

Q4. The process of taking in solid material by cell membrane is:

(A) Pinocytosis
(B) Exocytosis
(C) Phagocytosis
(D) Autophagy

Q5. Small pox is caused by:

(A) Bacteria
(B) Fungi
(C) Protozoa
(D) Virus

Q6. Which structure of bacteria helps in DNA replication?

(A) Mesosome
(B) Nucleoid
(C) Plasmid
(D) Cyst

Q7. Tests of actinopods are made up of:

(A) Calcium
(B) Potassium
(C) Silica
(D) Sodium

Q8. The fungi which obtain food from dead organic matter are:

(A) Autotrophs
(B) Saprotrophs
(C) Heterotrophs
(D) Parasites

Q9. In spermatophytes seed is formed from:

(A) Anther
(B) Embryo sac
(C) Ovary
(D) Ovule

Q10. Larva produced during the life cycle of annelids is named as:

(A) Trochophore
(B) Tadpole
(C) Bipinaria
(D) Brachiolaria

Q11. The hind limb of birds is modified for:

(A) Flying
(B) Running
(C) Perching
(D) Walking

Q12. During respiratory chain co-enzyme Q is oxidized by:

(A) Cytochrome – a
(B) Cytochrome – b
(C) Cytochrome – c
(D) Cytochrome – a_3

Q13. Acetic acid on entering mitochondria combines with co-enzyme – A to form:

(A) Malate
(B) Oxaloacetate
(C) Acetyl – CoA
(D) Fumarate

Q14. The term employed to the loss of appetite due to the fear of becoming obese is:

(A) Anorexia nervosa
(B) Bulimia nervosa
(C) Obesity
(D) Botulism

Q15. Breakdown of alveoli of lung is termed as:

(A) Asthma
(B) Tuberculosis
(C) Lung cancer
(D) Emphysema

Q16. Histamine that participate in allergic reactions is produced by:

(A) Monocytes
(B) Eosinophils
(C) Neutrophils
(D) Basophils

Q17. Antiserum is a serum containing:

(A) Antigen
(B) Antibodies
(C) Hormones
(D) Enzyme

BIOLOGY PAPER-I GROUP-I (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

> Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) How fats differ from oils?

(ii) Define an enzyme. Write names of parts of active two sites in enzyme.

(iii) Write any two characteristics of enzymes.

(iv) What is Induce Fit Model? Who proposed it?

(v) Differentiate the obligate and facultative parasite in fungi.

(vi) In what way composition of cell wall is advantageous to fungi with reference to nutrition?

(vii) How would you find contrast between ostia and osculum?

(viii) Write the functions of mantle and radula.

(ix) Echinoderms are comparatively simple organisms but are placed at the top of invertebrate phyla very close to chordates. Give any two reasons.

(x) What is Larynx? Give its function.

(xi) Define accessory pigments. What is their role?

(xii) How photophosphorylation differs from oxidative phosphorylation?

Q3. (Marks 16) Attempt any eight parts.

(i) What is meant by Phyletic lineage?

(ii) How would you differentiate deductive and inductive reasoning?

(iii) Give the role of endoplasmic reticulum.

(iv) What are cisternae?

(v) Write important features of diatoms.

(vi) Give ecological importance of dinoflagellates.

(vii) What are symptoms of malaria?

(viii) Differentiate foraminiferans and actinopods.

(ix) What is the affect of pH on capacity of haemoglobin to combine with oxygen?

(x) Give causes and symptoms of tuberculosis.

(xi) Write two functions of Monocytes.

(xii) How would you define source and sink?

Q4. (Marks 12) Attempt any six parts.

(i) What is prophage? How it differs from virion?

(ii) Differentiate slime and endospore.

(iii) Write distinguishing characters of bryophytes.

(iv) Write the structure of ovule of angiosperms.

(v) Write two differences between monocots and dicots.

(vi) What is pollen tube? Write its function.

(vii) What are detritivores? Give an example.

(viii) Define peristalsis and antiperistalsis.

(ix) What is chyme? Give its effect on duodenum.

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [4] How biology is helpful for protection and conservation of environment?

(b) [4] In what way respiration in birds is the most efficient and elaborate?

Q6.

(a) [1+3=4] Draw the structure of a Mononucleotide. Differentiate DNA and RNA.

(b) [1+3=4] What is the importance of unicellular fungi? Discuss ecological impact of fungi.

Q7.

(a) [1+1+1+1=4] Write any four differences between Prokaryotes and Eukaryotes.

(b) [1+1+1+1=4] Discuss food selection, grinding, lubrication and digestion functions of oral cavity of man.

Q8.

(a) [1+3=4] What is Hepatitis? Describe its different types.

(b) [4] Write down any eight functions of blood.

Q9.

(a) [4] Classify bacteria with respect to flagella.

(b) [4] Sketch the phases of glycolysis.

GROUP – II

BIOLOGY PAPER-I GROUP-II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 2462

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question.

Q1. The reasoning that moves from general to specific is called as:

(A) Deductive
(B) Inductive
(C) Scientific
(D) Theoretical

Q2. Animals mainly obtain carbohydrates from:

(A) Glucose
(B) Glycogen
(C) Sucrose
(D) Starch

Q3. Metal ions are related to:

(A) Co-enzyme
(B) Co-factor
(C) Vitamin
(D) Substrate

Q4. Which is not found in secondary wall?

(A) Chitin
(B) Inorganic salts
(C) Cutin
(D) Silica

Q5. The process in which viral DNA becomes incorporated into the bacterial chromosome is known as:

(A) Induction
(B) Lysis
(C) Lysogeny
(D) Deduction

Q6. Rapid phase of growth of bacteria is called:

(A) Log phase
(B) Lag phase
(C) Decline phase
(D) Stationary phase

Q7. Parasitic protozoans that form spores at some stage in their life cycle belong to:

(A) Actinopods
(B) Ciliates
(C) Zooflagellates
(D) Apicomplexans

Q8. Asexual reproduction in yeast occurs by:

(A) Conjugation
(B) Budding
(C) Fragmentation
(D) Conidia

Q9. The class of seedless plants containing foliar sporangia is:

(A) Angiospermae
(B) Gymnospermae
(C) Filicineae
(D) Algae

Q10. The largest invertebrate animal is:

(A) Anodonta
(B) Oyster
(C) Octopus
(D) Squid

Q11. Which of the given has a pseudocoelom?

(A) Ascaris
(B) Earth worm
(C) Hydra
(D) Planaria

Q12. Ferredoxin is a protein that contains:

(A) Copper
(B) Iron
(C) Magnesium
(D) Sodium

Q13. The NADH molecule provides the reducing power for the synthesis of sugar during:

(A) Chemiosmosis
(B) Electron transport chain
(C) Calvin cycle
(D) Glycolysis

Q14. Which type of muscles are found in stomach?

(A) Skeletal
(B) Smooth
(C) Cardiac
(D) Voluntary

Q15. Blood is not involved in exchange of gases in:

(A) Fish
(B) Frog
(C) Man
(D) Cockroach

Q16. Guttation occurs in plants through:

(A) Hydathodes
(B) Stomata
(C) Cuticle
(D) Lenticels

Q17. The type of white blood cells which perform Phagocytosis in tissue are:

(A) Basophils
(B) Eosinophils
(C) Monocytes
(D) Neutrophils

BIOLOGY PAPER-I GROUP-II (Subjective Type)

Time Allowed: 2.40 Hours | **Maximum Marks:** 68

Note: Write same question number and its parts number on answer book, as given in the question paper.

Q. SECTION-I

Q2. (Marks 16) Attempt any eight parts.

(i) Give one similarity and one difference between Amylose and Amylopectin.

(ii) What are enzymes? How they accelerate a metabolic reaction?

(iii) Write the effect of temperature on the enzyme action.

(iv) How would you differentiate activator and co-enzyme?

(v) How mycorrhizal association increases growth of plants?

(vi) What are toad stools? Give two examples.

(vii) Why annelids and arthropods are considered having same origin?

(viii) How would you differentiate ostia and osculum?

(ix) Write down the economic importance of Molluscs.

(x) Differentiate the determinate and indeterminate cleavage.

(xi) Absorption and action spectrum are different. How?

(xii) Name the processes, which acts as energy-capturing and energy releasing.

Q3. (Marks 16) Attempt any eight parts.

(i) How would you recognize a living organism?

(ii) Define community with an example.

(iii) How polysomes are formed?

(iv) What role is played by centrioles in cell division?

(v) How protista are different from prokaryotes?

(vi) How algae differ from the plants in sex organs?

(vii) What do you know about kelps?

(viii) How slime molds survive during unfavourable conditions?

(ix) If photorespiration is inhibited even then plants can grow. Then why does photorespiration exists?

(x) How counter current exchange increases amount of oxygen in birds?

(xi) How Abscisic acid controls stomatal movement in plants?

(xii) Why transpiration is called a necessary evil?

Q4. (Marks 12) Attempt any six parts.

(i) What do you know about capsid and capsomeres?

(ii) What are pili? Give their functions.

(iii) How "venus fly trap" catches and digest the insects?

(iv) Differentiate nutrients and nutrition.

(v) How would you define detritivores? Give one example of detritivore animal.

(vi) Funeria is an "amphibians of plant." How?

(vii) What is phylogenetic system classification?

(viii) Give two important features of female cone of pinus.

(ix) What are sori? Give their structure.

Q. SECTION-II

Q. (Marks 24) NOTE: Attempt any three questions.

Q5.

(a) [2+2=4] Give the role of Biology in the field of protection and conservation of environment.

(b) [3+1=4] Compare the role of haemoglobin and myoglobin in respiration.

Q6.

(a) [4] Why carbon is called the skeleton of life? Justify it.

(b) [1+3=4] Enlist different modes of nutrition in fungi. Describe fungi as predators.

Q7.

(a) [4] Describe structure and functions of lysosomes.

(b) [4] Why is digestion necessary? Describe what happens to a meal containing fats, carbohydrates and protein while it is in stomach of man.

Q8.

(a) [2+2=4] Write characteristics and structure of viruses.

(b) [4] How ascent of SAP takes place in plants? Explain only Cohesion Tension Theory.

Q9.

(a) [2+2=4] Explain the germ theory of disease. Also describe the discovery of bacteria.

(b) [4] Draw outline of glycolysis. No description is required.